



Sensors for daily life: A review

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ABSTRACT

Sensor technologies have improved the everyday life of human beings through their applications in almost all fields. Sensors are devices that detect changes in the source/environment and collect signals, and accordingly, the reaction is designed. There is a range of sources, including light, temperature, movements, and pressure etc., which may be used. A wide range of applications is utilised using innovative sensor technologies in lifestyle, healthcare, fitness, manufacturing, and daily life. In the medical field, the difficulty to take medicine is eased by drug donors fitted with sensors. It reminds them to take medicine via a signal and also supply the necessary medicine at the specified moment. In health care, older individuals, athletes, and risk patients benefit from modern sensor technology. The current industrial trends driving innovation include ultrasound, radar, and non-contact optoelectronic solutions and laser technology. The paper gives a brief overview of the numerous types of sensors that are utilised in everyday life. Various capabilities of sensors for day-to-day healthcare are discussed. Various features, associated nomenclature, and measures for sensors in day-to-day routine life are discussed diagrammatically and finally, the paper identifies and discusses twenty-two significant applications of sensors for daily life. Sensors also produce vital information and exchange data with other connected devices and administration systems when linked to a network. Thus, for the effective running of many companies, sensors are critical. Various types of sensors are used in our daily life, which is more accurate and makes quicker analysis.

1. Introduction

Sensors capture and translate their physical attributes into observable electrical impulses from chosen surroundings. Temperature, mass, speed, pressure, or heat bodies such as people are included as these attributes. A microprocessor processes the electrical impulses to provide outputs that correspond to a set of measures. The system sends the output to the recipients in the designated devices. A system can use many sensors with varied capacities, depending on functional complexity and increasing functional requirements [1–3]. Sensors increase the capability of the world around us to observe and report. They are made to work to make human lives considerably more accessible and better in nearly all fields. Setting up moods, switching on water heaters, guaranteeing safety, tracking equipment, and more are some of them. Sensors allow better

visibility incorporate processes and workflows, analyse patterns of employees' work and detect environmental conditions in facilities on a larger scale. These can monitor, regulate, and increase operational efficiency in business management [4,5].

Electrical sensors transform a stimulus into an electrical signal, then processed by the computer into meaningful end-user information. Sensors for medicine are one of the most complex technologies to design and reliably integrate with Smartphones and the Internet of Things (IoT) with the necessary capabilities. Biological sensors employ biological molecules to detect particular objective chemicals as receptors [6,7]. These key technological elements monitor the heat that an object or system releases. These allow us to feel a temperature shift bodily. Prevention is a crucial function of temperature sensors. When a predefined high point occurs, temperature sensors detect time for preventive action [8–10].

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Today, the world has advanced so far that many key processes would not be restored without sensor data. These are used in our homes, at our shopping centre's, and our hospitals. They are included in Smartphone's and are part of the IoT. Cost reduction and a dramatic improvement in guest experience are achieved with sensors built for smart hotels. In automating temperature controls and light settings, the thermostats and occupancy sensors offer smart energy management, making sensible energy use [11–14]. Sensor-based technologies change the way people connect and work in today's workplace. This change involves intelligent sensors, which provide enhanced productivity and performance [15–17].

Since sensors are far smaller than hair in textiles, the wearer of the garment is rarely noticeable. A new generation of tissue-integrated sensors can monitor innovatively biological happenings. Biosensors are taking vital monitoring of daily life activities of the health. These biosensors are not isolated from the body but function entirely inside the body without metal and electronic components. This technology provides additional security and relief in everyday life, particularly for chronically sick people. It can be mounted on the upper arm and enables a discrete blood glucose measurement [18–20]. This study shows various sensors and their capacities for day-to-day healthcare. The major strength of this paper is to identify and discuss various applications of sensors for daily life.

2. What are sensors?

A sensor is a device, module, machine, or subsystem that detects events or changes in its environment and relays the information to other electronics, most commonly a computer processor. A sensor converts physical phenomena into a measurable digital signal, which can then be displayed, read, or processed further. The figure illustrates the working of a sensor. Various specialists and researchers classify sensors in a variety of ways. In the first classification, the sensors are divided into Active and Passive categories. To work, active sensors need an external excitation signal or a power signal.

On the other hand, passive sensors do not require any external power and produce an output response. GPS and radar are examples of active sensors that require an external power source to operate. Active remote sensing techniques such as RADAR and LiDAR measure the time delay between emission and return to determine an object's location, speed, and direction. Passive sensors, also known as self-generated sensors, produce their own electric signal and do not require external power. Thermal sensors, electric field sensing, and metal detection are examples of these. The sensor's detecting method is used in the other categorisation method. Detection methods include electric, biological, chemical, radioactive, and other methods. Another classification is based on conversion phenomena, such as input and output. Thermoelectric, Photoelectric, Electrochemical, Electromagnetic, Thermo-optic, and other common conversion processes are only a few examples [21–24]. The illustration view on the working of sensor is shown in Fig. 1.

3. Some key sensors for daily life

We live in an information age when we expect to know everything right away – and to be able to access information from anywhere, at any time. Sensor technology can be used in various ways in everyday life, particularly in smart homes, ranging from sensor-controlled burglary and fire prevention to heating and lighting management to modern household control. Cleaning robots can clean a house on their own, regardless of the type of flooring. Sensors and cameras guarantee that the vacuum cleaner cleans the house from every aspect and avoids impediments. Wiper and window cleaning robots, as well as autonomous lawnmowers, operate on the same concept. Water sensors can help prevent future water damage from washing machines and dishwashers in the home. Sensors will continue to grow into every part of our life as technology advances. Sensors are used worldwide to improve transportation, medical treatment, nanotechnology, mobile devices, virtual and augmented reality, and artificial intelligence (AI).

Sensors have become an indispensable part of modern living. If you're reading this on a computer, you're probably certainly using an optical mouse. Touch sensors detect every time a touch is made on a Smartphone's screen. Sensors come in a variety of shapes and sizes. There are dozens of different types of sensors in the average automobile. Tire pressure sensors determine if a tire is flat or needs to be inflated. Self-driving vehicles, such as the Tesla, include ultrasonic sensors that use sound waves to estimate the distance between the vehicle and other objects in its environment. Motion sensors are used in home security systems to detect the movement of bigger items. The most often used motion sensor for home monitoring is a Passive Infrared (PIR) device, which detects infrared radiation in the sensor's surroundings. Medical gadgets commonly employ sensor technologies. In prosthetic technology, input sensors such as myo-electrodes are employed. Electrical signals from a patient's muscular contractions are detected using myo-electrodes. Patients' pulses are monitored and detected by heartbeat sensors, while thermometers measure the temperature. Sensors are embedded in everything we come into contact with in our everyday lives. Subsequent subsections show some of the critical sensors used in our daily life are discussed under:

3.1. Level sensors

A sensor used to determine the level or amount of fluids, liquids, or other substances flowing in an open or closed system is referred to as a “level sensor” [25,26]. These sensors may be found in a variety of sectors. They are most recognised for gauging fuel levels, but they are also employed in industries that deal with liquids. The recycling business and the juice and alcohol industries rely on these sensors to track their liquid assets [27,28].

Some of the best use cases for level sensors include fuel gauging and liquid levels in open or closed containers, sea level monitoring and

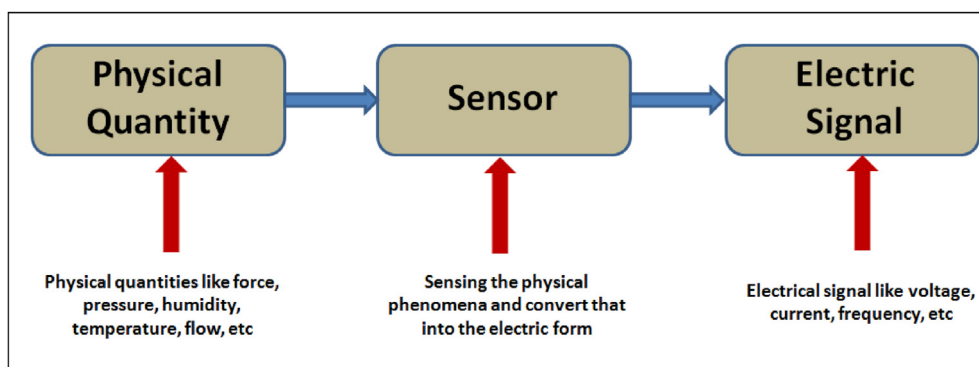


Fig. 1. Illustration view on the working of sensor.

Tsunami warning, water reservoirs, medical equipment, compressors, hydraulic reservoirs, machine tools, beverage and pharmaceutical processing, high or low-level detection, and so on [28–30]. Sensors capture all of the necessary data at all times, which helps them simplify their operations. Any product manager may utilise these sensors to know exactly how much liquid is ready to be delivered and whether production should be increased.

3.2. Temperature sensors

Temperature sensors were primarily used for air conditioning control, freezers, and other environmental control devices. Now they are used in manufacturing, agriculture, and the healthcare sectors. Because a defined ambient temperature and device temperature are required for much of the equipment in the manufacturing process, this type of measurement can always be used to improve the production process. The soil temperature, on the other hand, is crucial for crop growth in agriculture. It helps plants develop correctly, allowing for optimal output [31–35].

3.3. Proximity sensor

Proximity sensors are commonly utilised in the retail business since they can detect motion. Vehicles are another important and long-standing use case. The proximity sensor alerts automobile drivers while reversing for any time of obstruction and GPS command. They're also used to figure out how much parking there is in malls, stadiums, and airports [36–40].

3.4. Pressure sensor

Liquid or other forms of pressure are used in a variety of devices. These sensors enable the creation of IoT systems that monitor pressure-driven systems and devices. Any variation from the typical pressure range alerts the system administrator to any issues that need to be addressed [41–43]. The use of these sensors is beneficial not only in production but also in the maintenance of complete water and heating systems since it is simple to detect any pressure fluctuations or decreases [44–46].

3.5. Water quality sensor

Water is utilised almost everywhere. These sensors are critical because they monitor water quality for a variety of uses. They are employed in a wide range of sectors. Water quality sensors are used in water distribution systems for several reasons. Contamination from non-potable water cross-connections, polluted water entering the distribution system through leaking pipes in a low-pressure location, or microbial growth in distribution system pipes is all issues that need to be addressed [47–50].

3.6. Chemical sensor

Chemical sensors are used in a wide range of industries. Their purpose is to detect liquid changes or chemical changes in the air. In larger cities, where it is required to watch developments and safeguard the populace, they perform a vital role [51,52]. Chemical sensors are used in various applications, including industrial environmental monitoring and process control, detecting harmful chemicals released intentionally or inadvertently, explosive and radioactive detection, recycling processes on the International Space Station, pharma industries, and laboratories [53,54].

3.7. Gas sensor

Gas sensors are similar to chemical sensors, except that they monitor air quality and detect various gases. They are used for air quality monitoring, toxic or combustible gas detection, and hazardous gas monitoring

in coal mines, oil and gas industries, chemical laboratory research, and manufacturing – paints, plastics, rubber, pharmaceutical and petrochemical, and related products [55–57].

3.8. Smoke sensor

A smoke sensor detects smoke (airborne particles and gases) as well as its amount. They have been around for quite some time. They are now even more effective, thanks to the advent of IoT because they are hooked into a system that quickly warns the user of any problems that arise in various businesses. Smoke sensors are widely employed in the industrial business, HVAC, buildings, and lodging accommodations to identify safety hazards. It serves to safeguard those working in hazardous areas, as the system as a whole is far more effective than previous ones [58–60].

3.9. Infrared (IR) sensors

An infrared sensor emits or detects infrared radiation to perceive particular features of its surroundings. It can also detect and measure the heat radiated by the items. They are currently being employed in a range of IoT projects, particularly in healthcare, because they make blood flow and blood pressure monitoring straightforward. They are also found in various other smart gadgets, like smartwatches and smartphones [61, 62].

Examples of typical applications include home appliances and remote control, breath analysis, infrared vision (to visualise heat leaks in electronics, monitor blood flow, and allow art historians to see beneath layers of paint), wearable electronics, optical communication, non-contact-based temperature measurements, and automotive blind-angle detection [63–65]. They are helpful for more than just that; they are also terrific for ensuring high-level security in the house. They can also detect a range of chemicals and heat leakage; therefore, their use covers environmental monitoring. They will play a significant part in the smart home market due to their many uses.

3.10. Image sensors

Image sensors are electronic devices that convert optical images into electrical signals that can then be displayed or stored. Image sensor applications include digital cameras and modules, medical imaging and night vision equipment, radar, thermal imaging devices, sonar, biometric, IRIS systems etc. [66–68].

3.11. Motion detection sensors

A motion detector is an electrical device that detects physical movement in a particular area and converts it into an electric signal; it may detect the movement of any item or human person. The security business relies heavily on motion detection. These sensors are used in regions where no movement should be observed at all times, and they make it simple to notice anyone's presence when installed. Intrusion detection systems, automatic door control, boom barrier, smart camera (i.e., motion-based capture/video recording), toll plaza, automatic parking systems, automated sinks/toilet flushers, hand dryers, and energy management systems are among the most common applications such as Automated Lighting, AC, Fan, Appliances Control, etc. [69–72].

3.12. Accelerometer sensors

An accelerometer is a type of transducer that converts mechanical motion into electrical signals by measuring the actual or quantifiable acceleration that an item experiences due to inertial forces. It is the rate at which velocity varies concerning time. These sensors are now found in millions of goods, including smartphones. Vibration sensing, tilting, and acceleration are only a few of their uses. It's perfect for keeping track of a fleet of vehicles or using a smart pedometer [73–75].

They're used in cellular and media devices, vibration measurement, car control and detection, free-fall detection, aircraft and aviation sectors, movement detection, sports academy/athlete behaviour monitoring, consumer electronics, industrial & construction sites, and other places [76,77].

3.13. Gyroscope (gyro) sensors

Sensors that monitor angular rate or angular velocity are known as gyro sensors. A measurement of rotating speed around an axis is known as angular velocity. It is mostly used for navigation and angular and rotational velocity measurement in three axes. The most important application is tracking an object's orientation. Among their most typical uses are car navigation systems, gaming controllers, cellphone and camera devices, consumer electronics, robotics control, drone & RC control helicopter or UAV control, vehicle control/ADAS, and many others [78,79].

3.14. Humidity sensors

The quantity of water vapour in an environment of air or other gases is known as humidity. "Relative Humidity" is the most widely used phrase (RH). Their applications and use may be found in the industrial and residential domains for controlling heating, ventilation, and air conditioning systems. They are also used to preserve pharmaceuticals in automobiles, museums, industrial areas, greenhouses, meteorological stations, paint and coatings businesses, hospitals, and the pharmaceutical industry [80–84].

3.15. Optical sensors

An optical sensor is a device that detects the physical amount of light rays and converts it to an electrical signal that a human or an electronic instrument/device can read. As a result, these sensors are found in various industries, including healthcare, environmental monitoring, energy, aerospace, and many more. Oil companies, pharmaceutical companies, and mining companies are considerably better able to track environmental changes while simultaneously protecting the safety of their employees. Ambient light detection, digital optical switches, optical fibre communications, high-speed network systems, elevator door management, assembly line part counting, and safety systems are only a few of their principal applications. They're ideal for oil and gas applications, civil and transportation domains, high-speed network systems, elevator door management, assembly line part counting, and safety systems because of their electrical isolation [85,86].

4. Sensors for day-to-day healthcare

Today, biosensors are very much used for healthcare in day-to-day activities. A biosensor is a device that allows biological reactions, such as heart rate or brain waves, to be detected and converted to quantifiable information. Biosensors can collect biological signals and transform them into comprehensible, workable data using biometric algorithms [87–90]. Biosensors are biological devices that are utilised in sensing and include a physicochemical detector. This technology powers a large proportion of the mobile and wearable device market. It enables users to detect everything from heart and breathe to stress and tiredness using mobile wearable devices. Biosensor technology has also significantly improved the monitoring of glucose levels for the common man. This technology can significantly influence the health business and the daily lives of all those who utilise it [91,92]. Bio-sensors can deliver economic, user-friendly, sensitive, and highly accurate detection instruments [93, 94].

The devices made from them are suitable for various applications, from clinical to environmental and agricultural applications. Currently, wearables are designed to monitor several health characteristics such as

blood pressure, heart rate, and illness symptoms. For instance, high blood pressure; numerous garments with biosensors were produced with systematic and diastolic blood pressure data collected. It enables the user to recognise triggers simply and report to the doctor to establish an efficient treatment and care plan [95,96]. Biosensors are used in a wide range of applications for medical and clinical processes. This is helpful for screening and diagnosis of disease, research and commercial applications. It is now cost-effective, sensitive and highly accurate detection devices are now available. These devices detect the amount of blood glucose, track body reactions, illness, and monitor few types of patients.

5. Features and typical markets for sensors in day-to-day routine

Fig. 2 is exemplifying the various traits and detailed associated solicitations of the sensors employed in routine life. There have been reported umpteen sectors of daily span namely; automotive, power, aerospace, beverages & food, chemical & petrochemicals, life sciences, etc., where the highlighted sensors have been practised throughout the several industrial sectors [97,98].

Sensors are available in numerous applications and situations. In the development of driverless automobiles, optical sensors play a significant role. Optical sensors, including breath analysis and cardiac monitoring, are also utilised in the biomedical industry. These systems monitor heart rate, consumption of calories, sleep, the activity of tracking, and many other measurements to keep us healthy. In certain situations, these wearables can contact other applications and exchange information with a health care practitioner on users' chronic diseases. Besides the individual usage of health wearables, clever advanced appliances like scales, thermometers, blood pressure monitors, and even hairbrushes are now available. The mobile application links to a connected device to operate tasks such as door opening/closing, engine metrics, alarm, position, traffic detection, etc. [99–101].

Sensors execute several crucial measures to Industrial Internet of Things (IIoT) functioning, such as monitoring processes, measuring measures, and data collection. For inventory management and process control, real-time level measurement in big tanks, containers, and vessels is necessary. Level sensors are among the most frequently accessible intelligent sensors. They can be utilised for tanks, diesel fuel, trash management, recycling, and irrigation measurement. Water treatment and rainfall quality may be monitored via water sensors. They are also beneficial for aquariums and aquatic theme parks. In particularly demanding situations, such as environmental control rooms and high-tension pipes, specialised sensors have also been developed. Various moisture sensors have also been designed to measure the water dissolved in oils, which helps preserve our energy infrastructure [102–104]. Critical to monitoring, detecting, and assuring safety, air quality, and environmental analysis in a wide array of sectors are the gas sensors.

6. Associated nomenclature and measures for sensors in routine life

Fig. 3 is reflecting about the relevant measures and nomenclature allied with the sensors in daily life. The significant properties that have been revealed for the daily life sensors are namely; altitude, flow, distance, pressure, sound, torque, tilt, conductivity, temperature, etc. These sensors and their pertinent capabilities have further enabled these sensors to be more prominent and advantageous for specific engineering sectors [105,106].

In the medical and life science sectors, use large numbers of gas sensing devices. Due to the exact, dependable, and constant gas sensors, they can be integrated easily into sectors such as incubation, neurology, breathing, capnography, etc. Aerospace uses gas sensors, especially for in-flight monitoring and air quality management, to guarantee staff productivity and overall passenger convenience. Many clients regularly search for the gas sensor technologies typically used in industrial procedures in hostile and robust surroundings. It is used with certain

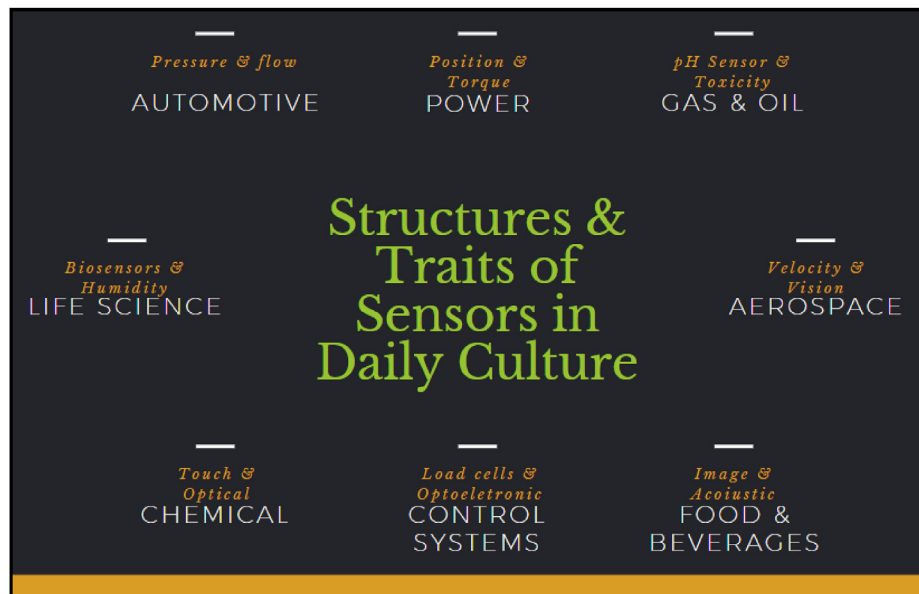


Fig. 2. Several traits and typical market services of sensors.

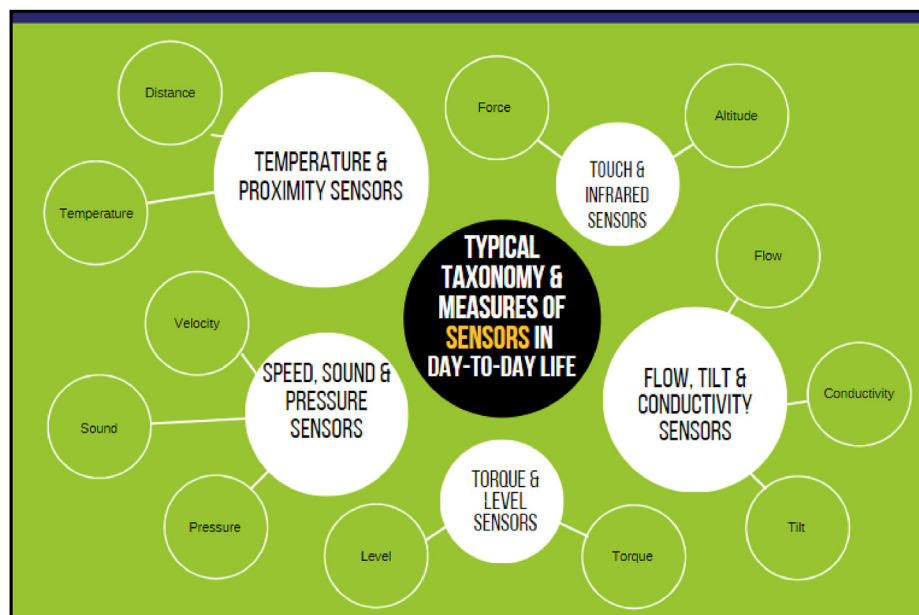


Fig. 3. Taxonomy and relevant measures of sensors for daily span.

complex circumstances and be environmentally sustainable while dealing with extreme environmental situations and industrial gases [107, 108].

The sensors sense the elements and happenings in an area through the capture of sound waves.

Many sensors are used concurrently by AI-enabled devices and apps. Manufacturers can improve accuracy, maintain a competitive edge and concentrate on innovation by leveraging complementary data sources. This can be achieved when ML algorithms are predictive and the data engineering procedures. Connected devices can collect medical and other health data using an intelligent medical gadget connected to a Smartphone app. In many applications, algorithms can handle visual sensor data. Security systems monitor live movements, track faces, and recognise the identity. Similarly, self-driving automobiles evaluate in real-time trajectory and predict pedestrian intents. Innovation and development prospects for more advanced software products include

algorithms and visual sensing technologies [109–111].

7. Sensors applications for daily life

Sensors are independent devices integrated into regular items or machines for smart use. This allows measuring the noise level in a particular setting. In smart city solutions, acoustic sensor systems can detect and deliver data to avoid noise pollution. The most common application of sensors can conceive as a smart building system. This technology can continuously monitor and records physiological and exterior data is as personal as sensor technology [112–116]. Temperature sensors can be utilised in almost every setting, probably the most adaptable smart sensor. These can monitor, measure and warn an operator or an emergency shut-off system of the temperature of a machine in an industrial context. This technology can measure and follow blood flow during various medical processes, monitor buildings and structures' heat

Table 1
Sensors applications for daily life.

S. No	Applications	Description
1	Managing day to day activities	There are many specialised applications of sensors devices in day-to-day activities. These are used to manage key processes in the chemical, oil, gas, and nuclear energy sectors. Here, sensors are required to control geographically vast power transmission and distribution systems in the electric energy business as well. Standards for communication, interoperability, dependability, calibrating, safety, and accuracy for intelligent sensors have been created for such sorts of vital infrastructure.
2	Safety	Devices can also be networked in conjunction with the Internet and can be used for enhanced safety and convenience through intelligent home solutions. These solutions involve networks of data collection, cloud, and machine-to-machine communication. The sensor offers more data about the path of glucose than previously. With distance sensors and a processing unit, the warning system incorporated into the shoe delivers audio, haptic, or visual input. It permits free and ambient events on natural auditory channels. It is revolutionary for the standard fall mats to be supplemented with fall mat sensors that rapidly record load changes and ask for assistance via the call system.
3	Cleaning	Sensors guarantee that the flat is cleaned at every aspect and vacuum cleaners can avoid physical barriers. In the future, water sensors can also prevent damage to the house caused by washing machines or dishwashers. Smart sensor systems are everywhere in our daily lives. They offer safety, save lives and enhance living quality. More and more automated and networked living sectors will raise the necessity of novel sensor technology.
4	Emergency	Driver assistance systems like autonomous emergency braking systems or parking devices facilitate daily life and boost road safety via cameras, infrared and ultrasonic sensors, and radar sensors. This offered automatic parking aids using ultrasonic front and rear sensors to support the guides during parking. These help to identify parking spaces available and relay information. Free parking places can be accessed immediately by drivers. The automobile lacks the parking area, the wheel locks itself, and only the throttle and brake have to be operated by the driver. Innovative, self-sufficient energy parking sensors facilitate daily living to locate a place in urban areas.
5	Food quality control	Sensors in the food sector used for quality control operations are utilised for measuring carbohydrates, alcohol, and acids. These devices are also being employed to monitor fermentation. The use of pathogens in fresh meat, poultry, or fish is very vital to identify. For air and water quality control, sensors are successfully utilised. It can be used to collect traces of pesticides from organophosphates and to monitor wastewater toxicity levels. A kettle utilises temperature sensors to guarantee the temperature reaches the proper degree while preparing a cup of coffee. A coffee maker utilises pressure switches to check the volume of liquid delivered.
6	Boilers and washing machines	Some household appliance applications using sensors are the boiler, washing machines and dishwashers for heating systems, and other white products. Many vehicles, including automobiles, trains, buses, etc., employ sensors to monitor petroleum temperature and pressure, jets and steering systems, and so many more. Magnetic sensors are used to detect movement by the use of the magnetic field of the earth. It has several uses, such as portable navigation, product promotion, indoor and urban navigation, etc.
7	Measure distance	Sensors are commonly employed in all forms of portable and handheld devices. They are utilised in fitness applications to measure the distance of each step and the incremental steps that a person has taken. These are also used to detect the user's motion, such as rotating and tilting, providing them with an incredible gaming experience. Motion sensors also have a wide range of home and personal safety applications. However, without a motion sensor, it is difficult to establish a safety system. They are particularly successful because during peak night hours; they offer clear views.
8	Doorbells	Sensors play a significant role in wireless doorbells. It may dispose of typical entrance cells employing motion sensors, as movement sensors can inform guests when they reach the door. Motion sensors detect motion using optical, vibration, radiofrequency, sound, and warmth, essentially using electronic chips. The centre of our new safety systems is movement sensors. They are utilised for the detection and understanding of location, speed, and acceleration.
9	Automating lighting operations	The primary operation of light sensors is tuning light on the day. In addition to facilitating human-centric lighting, daylight harvesting, and lighting planning for energy savings, efficiency, and the environment, it automates lighting operations in smart buildings. Thermal sensors are required to automate room temperature in smart buildings and businesses. The temperature is continuously inside, regardless of how the temperature swings externally. These sensors also conserve energy by regulating the temperature with the climatic environment.
10	Assure home security	Sensors are designed to assure security for the house since they provide fast warnings and trigger fire burst alarms. They boost the possibility of escaping accident situations dramatically. In addition, more sensors such as time sensors, position sensors, sound sensors, material sensors, and many more are used for this purpose. It assists manufacturers to improve their operations and staff well-being.
11	Monitor temperature changes	In different kinds and sizes, sensors are available. Temperature sensors monitor a source's heat energy content to detect and convert temperature changes into data. Machinery used in production generally demands temperatures for the surroundings and device at specific values. Similarly, the soil temperature is an essential component in crop growth within agriculture. Some are purpose-built with numerous integrated individual sensors so that multiple data sources may be monitored and measured.
12	Identify persons/person entering a room	Sensors are meant to identify persons while entering or leaving a room. The sensors transmit a signal to the light system when motion is detected to initiate procedures that reduce the energy supply. It automatically switches lights and other in-house equipment on/off based on the occupancy of the allocated space. In commercial applications which deal with wider regions and people, they can substantially save energy. It is also commonly used in hospitals for monitoring compliance with hand hygiene.
13	Measure water vapour	The measurement of water vapour in the air or other gases is carried out using sensors. In both the industrial and residential fields, humidity sensors are typically found in heating, air-conditioning systems. These are used in many different sectors, including hospitals and weather stations, for weather reporting and prediction. A pressure sensor detects gas and fluid changes. The sensor detects various changes and transmits them to linked systems when the pressure changes. Leak tests that can be a result of degradation are everyday situations of usage. Pressure sensors are also essential to produce water systems since variations or pressure dips can be easily detected.
14	Collects radiation	The sensors collect radiation that the item or surrounding regions produce or reflect. The most prevalent radiation source recorded by passive sensors is sunlight reflection. For example, photography, infrared, and radiometers are examples of passive sensors. Passive sensors are employed more because the satellite imagery is of exceptional quality. The passive sensor is preferable in technological monitoring globally, such as multiple spectral and hyperspectral technologies.
15	Avoid injuries	Sensors enable people to be monitored around them to avoid injuries from falling, overwork, heavy machinery, etc. In combination with innovative cognitive capacities and external sources such as environment and weather, sensors have good potential for better management, good health, and security to transform our way of life and work today truly. Wearable technology uses the IoT to gather, integrate, and analyse sensory data. A sensor is utilised to detect, measure, and report a single variable in the actual world in these systems.
16	Transit data about the person status	Sensors capture and transmit data about a person's physical status in real-time, which can minimise emergency response time and save lives. This might be important for older patients. These can quickly analyse the elderly persons living alone and provides the timely treatment. These are used to detect and monitor the level of oxygen concentration. Oxygen sensors are used in this system to control and monitor the predicted oxygen delivery to the patient. Electrochemical oxygen sensing elements are typically used to measure the oxygen level in ambient air.
17	Detect level of liquid and other quantities	Sensors are used to detect items such as liquids, powders, and granular solids. Many sectors, including oil, water and drinks processing, and food production, employ high-level sensors. Trash management systems are typical because level sensors can monitor waste levels in

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Table 1 (continued)

S. No	Applications	Description
18	Monitoring environmental factors	waste containers or dumpsters. These sensors measure the presence of fuel or hazardous gases and detect changes in air quality. Mining, oil and gas, chemical research, and manufacturing are the industries adopting sensors. Intelligent sensors monitor several environmental factors, including humidity, air temperature, and soil quality. These utilise weather and predictions in real-time to develop an ideal watering schedule for the yard. It plays a vital role in digitalising health services that enhance clinics' operations and the results of the patients. Sensors help to monitor the condition of smart warehousing, guarantee a rapid check-in and check-out and simplify day-to-day operations. Companies are increasingly the applications of sensors in industrial mining.
19	Identify movements	Sensors are commonly employed to identify movements in retail contexts. A shop can send discounts or offers to their mobile phone using the vicinity of a possible consumer. In parking garages, larger venues like airports, malls, and stadiums may check availability via proximity sensors. A moisture detection sensor can offer ahead for home's risk due to frozen pipes or even a damaged water line. The sensors can notify of leaks in the house to resolve the problem rapidly.
20	Home items	Sensors are installed around heaters, washing machines, refrigerators, sinks, pumps, and any leaking water danger. These can notify regarding unwanted water and quickly inspect the situation at home. When visitors come and leave the house, this technology may even turn the lights on and off, door and windows. Door and window sensors are the first protection lines for home breakdowns; even when an invader breaks a window, some sensors detect. Sensors can notify the movement of our home and the opening or closing of doors or windows. Motion sensors become an additional eye pair, warning about undesired activities like sneaking out adolescents.
21	Saving electricity	Sensors are employed to save electricity by linking them to lighting or the thermostat to regulate the energy use in a building depending on the occupancy. Motion sensors may also be connected to the video, which allows notification that the sensor has been triggered and activates the video recording to collect incursion footage. A multi-sensor integrates many sensors into one unit. Sensors may help to construct a safe and intelligent house. This technological adoption rapidly catches up and even becomes an insurance incentive. It has certainly helped us build smarter and safer homes.
22	Gas detection and surveillance	Gas detection and surveillance are always strongly related to the safety of people in difficult circumstances when the presence of toxic gases is identified. Sensors can easily monitor and measure the amount of gas frequently to be greater than usual and ensure the safety of persons operating in and near dangerous gases. Indoor air quality is another significant business that employs gas sensor technology to monitor.

losses, and much more [117–121]. Table 1 discusses significant applications of sensors in daily life.

Sensors are one of the most regularly used environmental monitoring technologies. Sensors enable the acquisition of data from the environment in very little time [122–125]. These can be connected to the cloud using various communications and transport modes, such as mobile networks, satellite networks, Bluetooth, broad-based networks, low-energy wide-band networks, etc [126–130]. A steam valve is used to control engine speed efficiently in conjunction with the mechanical sensor and actuator. The industry has created suitable standards for process control and digital and smart sensor networks and wireless, introducing digital technology [131–134]. It allowed for the communication of a rich range of device status, diagnosis, time, and other data together with the digital monitoring signal [134–137].

8. Discussion on the study

Various sensor technology applications in everyday life, particularly in smart homes, range from sensor-controlled burglary and fire prevention through heating and illumination management to current domestic control. Sensors and cameras ensure the home is cleaned from all angles, and the vacuum cleaner can prevent barriers. The same applies to autonomous lawnmowers for washing and window cleaning. The requirement for sensors accurately depends on the use. Generally, in terms of voltage, current, resistance, and capacity, the sensors' output is electrical. They need to be carefully described to verify that the sensors comply with the standards to make mistake repairs. The test devices also need a broad range to guarantee that they cover all the output range of the sensors. Ultrasonic sensors are also employed as the nearness sensor, which sets an impediment to the threshold distance. It is an excellent instrument for distance measurement without touch with the object and is a practical approach for accurately measuring tiny distances. As the time gap between signals transmitted and received is proportionate to the distance from the object, a simple analysis of this data may show the sensor's removal from the item.

Data during the surveying of different systems and control instruments are measured with the sensors used in the aviation sector. These data are used to improve aircraft performance and design and to enhance inefficient flying operations. Sensors enable measuring the ground, vibration, and environmental conditions testing and offering the pilot valuable information to control normal operations and emergencies.

There are several uses, each with its particular properties, for liquid level sensors. The most characteristic use is to measure the tank level of liquids where the height of the tank is known. Temperature sensors are made up of two metal alloys that generate electricity. This is proportional to the thermocouple conductor temperature difference. Sensors improve computers' ability to interact with their surroundings. In the future water, sensors can also prevent harm to water at home from washing machines or dishwashing machines.

8.1. Future scope

Sensors in the future will be smarter and more accurate in their measurements system. This technology is more effective and efficient and assists in providing appropriate decision-making options. With developments in process automation, sensing technology improves, resulting in more capable and better-networked sensing equipment. Sensors are spreading across a wide range of applications toward connected society. In the future, this will be helpful in many cases as per the requirements. There is a wide range of smart sensor applications in several fields. Future sensors will be more intelligent and precise in their measurements. Digitisation has had a favourable impact on costs, quality, and production lead time, among other factors. As a result, there is a growing demand for modern smart sensors. Food safety and biological hazard detection, safety hazard detection and warning, environmental monitoring, health monitoring and medical diagnostics, industrial and aerospace applications, smart antennas, vehicles, and smart roads will benefit from these smart sensor technologies.

9. Conclusion

In business and our everyday life, sensors are everywhere. In industrial systems, process variables like temperature, pressure, level, flow rate, and turbidity are measured accurately. The electrical variables like the voltage, current, and frequency; mechanical variables like revolution, cycle number, positioning, the direction of movement, static and dynamic pressures, proximity; environmental variables, such as humidity, vibration, wind velocity, and direction are also measured by using these technologies. In various areas, advanced sensor networks are employed and are being developed constantly. Through AI technology, sensor-based gadgets have become more innovative; these gadgets can interact and generate independent behaviour. They are used for receiving

information, data such as temperature, pressure, picture, etc. Remote sensing is when the information is obtained, detected, analysed, the physical parameters of an area are monitored, and radiated energy is recorded and released without physical contact with the studied subject. Sensors create a digital representation of the world in which they work by gathering data from houses, buildings, and vehicles. These can monitor and inform about the drug level in humans and detect any anomalies in the body. Sensors can assist in preventing symptoms and offering early warning of illness.

Declaration of competing interest

None.

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